DustLedger — presenter card (Delhi zonal, 10 min)

Open: zonal-deck.pdf on your own laptop · backups B1–B6 after slide 10 · team fields on slide 1 filled in · this card printed.

#	Time	Slide	Land this line
1	0:00–0:20	Title	"Delhi stops every site, clean or dirty, because it can't tell them apart."
2	0:20–1:10	Restrictions	Nov 1 – Jan 31 restricted; Dec 10 – Jan 20 total ban; GRAP-3 16 days in 2025.
3	1:10–2:00	Monitors & fines	47 monitors at city scale; fines ₹1 lakh to ₹2 lakh by area; checklists, not dust.
4	2:00–3:00	Small source, blunt tool	Construction = 2-5% of ambient PM10 (IIT Kanpur); 1,378,199 workers on the rolls.
5	3:00–3:50	Tools fall short	Winter PM10 ~314 µg/m³: a fence sensor mostly measures Delhi.
6	3:50–5:10	DustLedger	Downwind − upwind = the site's dust; "inconclusive" never means clean.
7	5:10–5:50	Forecasting	Persistence 23.8 µg/m³; ML +4% at most → not our spike.
8	5:50–7:00	Attribution	95.8 → 69.1 → **57.4** µg/m³ — **SIMULATED site**, real background + wind.
9	7:00–7:50	Cost & power	Parts ₹60,711 (parts only); 10.2 Wh/day; 2.1× in December.
10	7:50–9:15	Proof plan & ask	Real-site trial before 1 Nov or a permitted site; else controlled dust trial. Ask: site · co-location · DPCC contact.
—	9:15–10:00	Reserve	Stop. Take questions.

Three numbers to remember: ₹60,711 of parts per site · 57.4 vs 69.4 µg/m³ (self-calibration vs weekly reference visits, simulated) · construction = 2-5% of Delhi's ambient PM10.

Never say: tamper-proof · "we will cut AQI by …" · "clean sites can keep working" (it's a proposal) · "no one else does this".

Fast answers (full list: qa-prep.md)

- *Why you, in the most crowded theme?* Others measure the city; we answer which site added the dust — honestly, with "inconclusive" when we can't tell.
- *Built or simulated?* Data pipeline and forecast benchmark: built, real data. Site + sensors: simulated. Ring + misting: Round 2.
- *Sensors aren't accurate for dust.* Agreed (CSTEP 2025: qualitative). We use differences, then verify with a dust test and reference co-location; inspectors still verify before fines.
- *Calm winds?* Below 1 m/s in 16.2% of winter hours (ERA5, a lower bound) → "inconclusive", never clean.
- *Only a few per cent of PM — why bother?* Because that small source gets a blanket ban; targeting dirty sites is fairer. No city-AQI claim.
- *Trial during the ban?* Before 1 Nov or a permitted site; otherwise a controlled dust-source trial on campus.